

Module number	Module name	Lecturer responsible for the course
BIW-BA-BIW1-09 BIW-D-BIW1-09	Constructive geometry and geodesy	Prof. i.geodaesie@tu-dresden.de
Qualification goals	Students have a structured spatial ability and are to produce and properly interpret technical drawings and CAD representations. They can visualize spatial objects and solve related tasks constructively. Students know the definitions of coordinate and reference systems and are proficient in the evaluation of surveying data. They will be able to apply the methods of position and height measurement for the production of construction documents. They will be able to establish the connection between construction planning and surveying with the aim of complying with the required accuracy parameters of the geometry of the structure.	
Contents	Contents are - Practical applications of constructive geometric procedures, - Reference systems, coordinate systems and height systems in geodesy, - Coordinate calculation, - Operation of surveying instruments for measuring and setting out, - Area and volume calculation and - Surveying in road construction.	
Teaching and learning methods	3 SWS lecture, 1 SWS exercise, self-study	
Requirements for participation	Abitur knowledge in mathematics and physics at basic course level is required.	
Usability	The module is a compulsory module in the civil engineering degree course. The module creates the prerequisites for the module Fundamentals of Technical Infrastructure. The module is a compulsory module in the Bachelor's degree program in Civil Engineering. The module creates the prerequisites for the module Fundamentals of Technical Infrastructure.	
Requirements for the awarding of credit points	The credit points are earned when the module examination has been passed. The module examination consists of a written examination lasting 180 minutes. A 20-hour exercise is required for the examination.	
Credit points and grades	5 credit points can earned through the module. The module grade corresponds to the grade of the examination.	
Frequency of the module	The module is offered every summer semester.	
Workload	The total workload is 150 hours.	
Duration of the module	The module comprises one semester.	